ZS-A5

Dark Matter & ε-Halo

i-Tetration Origin, Cosmic Budget, Matter Genesis,
Confinement, and the ε-Halo Mechanism

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Theme: Astrophysics [ZS-A] | Paper 5 of 6
Verification: 50 checks (47 computed, 1 structural, 2 declarative) | All PASS | Zero Free Parameters
Consolidated from internal Z-Spin Collaboration research notes up to v2.2.0

§0. Abstract

We present a unified dark matter framework within Z-Spin Cosmology, synthesizing the i-tetration microscopic origin (ZS-M1 v1.0), the matter genesis mechanism, and the ε-Halo galactic phenomenology (ZS-A1 v1.0). Three convergent routes yield $\Omega_m \sim 0.32$ from **A** = 35/437 alone:

(1) i-Tetration route: The unique stable fixed point $z^* = 0.4383 + 0.3606i$ of $z = i^z$ ($k_W = 0$ branch, PROVEN unique) yields $\eta_{topo} = |z^*|^2 = 0.3221$.

(2) Sector counting route: Face counting (ZS-F2 v1.0 §11): $\Omega_b = F(\text{cube})/Q^2 = 6/121$, $\Omega_c = F(\text{truncated icosahedron})/Q^2 = 32/121$, $\Omega_m = 38/121 = 0.3140$ (Planck: 0.3153, 0.41%).

(3) Duality route: $\Omega_\Lambda/\Omega_m = m_d/m_u = 2e^A = 2.1668$ (cosmic: 0.36%, quark: 0.31%). Scale-invariant across 40 orders of magnitude.

The ε-Halo mechanism (Goldstone θ-mode gradient energy $\rho \propto 1/r^2$) produces flat rotation curves with zero dark matter particles. The stability threshold $S = |z^*|\cdot(\pi/2) = 0.8915$ generates the three-generation limit ($k_{G,max} = 1.36$) and quark confinement (Tension = $S + \sqrt{3}\cdot\mathbf{A} = 1.030 > 1$). The phase budget $\varphi(35) = 24$ connects spatial ($\varphi(5) = 4 = 2Z$) and temporal ($\varphi(7) = 6 = 2X$) constraints, with capacity $C = \varphi(35)\cdot\eta_{topo} = 7.73$ (≈96.6% fill).

Verification: 50/50 PASS. Eight pre-registered falsification gates.

Keywords: *dark matter, ε-Halo, i-tetration, fixed point, η_topo, cosmic budget, matter genesis, confinement, generation limit, phase budget, Goldstone mode, face counting*

§0.1 Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem (k_W=0 uniqueness, φ(35)=24, Pinch identity).
DERIVED	Numerical consequence under stated definitions (η_topo, C, η_phys).

CONSISTENT	Definition-dependent mapping (φ(5)=2Z, φ(7)=2X).
TESTABLE	Quantitative prediction with falsification condition.
HONEST	Limitation or non-claim explicitly documented.

§1. Three Convergent Routes to Ω_m

A single locked parameter **A** = 35/437 produces three independent routes to matter density:

Route	Formula	Value	Obs (Planck)
i-Tetration	η_topo = $\|z^*\|^2$	0.3221	0.3153 (2.2%)
Face counting	(6+32)/121 = 38/121 [PRIMARY]	0.3140	0.3153 (0.41%)
Duality inverse	$1/(1 + 2e^A)$	0.3157	0.3153 (0.12%)

The i-tetration and slot counting routes agree to 0.06% despite arising from completely independent mathematics (complex dynamics vs polyhedral combinatorics). Face counting (38/121) is now PRIMARY, matching Planck to 0.41%.

[STATUS: **DERIVED** (i-tetration & face counting routes); **DERIVED-CONDITIONA**L (η_topo ↔ Ω_m(face) bridge via ZS-F2 v1.0 §11.8)]
ηtopo and Ωm(face) are independently computed. The 2.5% gap between them is no longer "under investigation": ZS-F2 v1.0 §11.8 (Spectral–Index Projection Theorem) provides a three-layer structural account — Layer 1 (Z_2 gauge projection, 2.43%, DERIVED via Boundary Mode Theorem 11.7 + Atiyah–Bott equivariant index), Layer 2 (Seeley–DeWitt a_2 correction $\Delta a_2/e \approx 0.0241$, DERIVED-CONDITIONAL via ZS-M6 v1.0 §4.3), Layer 3 (higher-order Seeley–DeWitt residual, OPEN, $|\varepsilon_higher|/Q^2 < 4\times10^{-4}$, pending ZS-F7 v1.0 §8.1 Heat Kernel Pipeline closure under Falsification Gate F-BMT2). Layers 1+2 account for 98% of the gap; Layer 3 upper bound covers the remainder.

§2. i-Tetration Fixed Point and η_topo

2.1 The Unique Stable Fixed Point

$$\mathbf{z = i^z \Rightarrow z^* = (2i/\pi)\, W_0(-i\pi/2) = 0.4383 + 0.3606i} \quad (1)$$

k_W	Re(z*)	Im(z*)	|z*|	|f'|	Stable?
0	0.4383	0.3606	0.5676	0.8915	✓ YES
1	−1.862	−0.411	1.907	2.995	✗ NO
2	−5.878	−1.139	5.988	9.405	✗ NO
≥3	...	...	>10	>15	✗ NO

[STATUS: **PROVEN**] k_W=0 uniquely attractive. $|f'(z^*)| = 0.8915 < 1$. All other branches repulsive.

2.2 Topological Threshold

$$\eta_{topo} \equiv |z^*|^2 = 0.3221 \quad (2)$$

$$S \equiv |z^*| \cdot (\pi/2) = 0.8915 \quad \text{(stability budget usage: 89.15\%)} \quad (3)$$

2.3 Gravitational Enhancement

$$\eta_{phys} = (1+A)\, \eta_{topo} = 1.0801 \times 0.3221 = 0.3479 \quad (4)$$

§3. Cosmic Budget from Face Counting

Table 2. Complete cosmic budget (face counting, ZS-F2 v1.0 §11).

Observable	Z-Spin Formula	Z-Spin	Planck	Error
Ω_b	F(cube)/Q² = 6/121	0.0496	0.0493	0.58%
Ω_c/Ω_b	F(trunc.ico.)/F(cube) = 32/6	5.333	5.364	0.57%
Ω_m	(6+32)/121 = 38/121	0.3140	0.3153	0.41%
Ω_Λ/Ω_m	$2e^A$	2.1668	2.1746	0.36%
m_d/m_u	$2e^A$	2.1668	2.16±0.08	0.31%

The duality $\Omega_\Lambda/\Omega_m = m_d/m_u = 2e^A$ connects cosmic scales (Gpc) to quark scales (fm) through the same mathematical expression, spanning 40 orders of magnitude.

Master equation confirmation: The sector-counting partition $\rho_X:\rho_Z:\rho_Y = 3:2:6$ is independently confirmed as the unique equilibrium distribution of the Pauli master equation with Fermi golden rule rates $W_{AB} = \dim(B)\cdot\mathbf{A}/Q$ (ZS-Q7 v1.0 §5.1, Theorem 3A). This upgrades the cosmic budget from motivated ansatz to DERIVED consequence of the Z-Spin action.

[STATUS: **DERIVED** / **DERIVED under A1**] All ratios from locked (Z,X,Y,Q,**A**). Zero adjustable parameters.

§4. ε-Halo Mechanism: Dark Matter Without Particles

The U(1) completion (ZS-F1 v1.0) decomposes the Z-field as $\Phi = |\Phi|e^{\{i\theta\}}$: frozen radial mode ($m_\rho \sim O(M_P)$) plus massless Goldstone angular mode θ.

$$\Box\theta = 0 \quad \text{(exact, Goldstone theorem: } \theta \text{ has no potential)} \quad (5)$$

$$\theta(r) = \ln(r/r_s)/L \Rightarrow \rho_\varepsilon = M_P^2/(2L^2r^2) \quad \text{(isothermal halo)} \quad (6)$$

$$M_\varepsilon(r) = (2\pi M_P^2/L^2) \times r \Rightarrow v(r) = \text{constant} \quad \text{(flat rotation curves)} \quad (7)$$

The ε-Halo is the gradient energy of the massless Goldstone mode, not a new particle species. No WIMP, no axion, no sterile neutrino. Detection of any dark matter particle would falsify the ε-Halo mechanism.

MOND scale: $a_0 = cH_0/Y = cH_0/6 = 1.09\times10^{-10}$ m/s² (9% below MOND empirical 1.2×10^{-10}). Factor 6 = Y locked, not fitted.

[STATUS: **DERIVED**] Goldstone θ-mode satisfies □θ=0 exactly. Isothermal halo and flat v(r) are structural consequences.

§5. Matter Genesis: Generation Limit and Confinement

5.1 Pinch/Load Identity

$$\mathbf{Pinch(A,\theta) = 4\sin(A/2)\sin(\theta/2)} \quad \text{(PROVEN)} \quad (8)$$

Small-**A**: Pinch ≈ 2**A** sin(θ/2) + O($\mathbf{A}^3$). For color triplet (θ = 120°): Pinch = 0.1387 (√3·**A** = 0.1387, 0.027% deviation).

5.2 Three-Generation Limit

$$\mathbf{Tension_f(k_G) = S + k_G \cdot A}, \quad k_{G,max} = (1-S)/\mathbf{A} = 1.355 \quad (9)$$

k_G = 0, 1 stable; k_G = 2 borderline/metastable; $k_G \geq 3$ forbidden (Tension > 1). This reproduces the three-generation structure of the Standard Model.

[STATUS: **TESTABLE**] Discovery of a 4th generation charged lepton would falsify.

5.3 Quark Confinement

$$\mathbf{Tension_q = S + \sqrt{3} \cdot A = 0.8915 + 0.1387 = 1.030 > 1} \quad (10)$$

Isolated quarks (color triplet, θ = 120°) exceed the tension threshold. Color-singlet composites cancel the color load (θ → 0), removing the overflow. Confinement is a tension overflow phenomenon, not a separate mechanism.

5.4 Quark Mass Ratio

Model-E (working hypothesis): $m_d/m_u = 2e^A = 2.1668$ (PDG: 2.16 ± 0.08, 0.31% agreement).

Model-P (FALSIFIED): $m_d/m_u = 2e^{\sqrt{3}\cdot A} = 2.298$ (too large, >5σ from PDG).

§6. Phase Budget and Capacity

$$\mathbf{\varphi(35) = \varphi(5) \times \varphi(7) = 4 \times 6 = 24} \quad \text{(PROVEN)} \quad (11)$$

Factor	Value	Mapping	Interpretation
φ(5) = 4	Spatial	2 × Z = 4	CONSISTENT

$\varphi(7) = 6$	Temporal	$2 \times X = 6$	CONSISTENT
$\varphi(35) = 24$	Total	$4 \times Y = 2 \times G$	PROVEN

$$\mathbf{C \equiv \varphi(35) \times \eta_{topo} = 24 \times 0.3221 = 7.73} \quad \text{(capacity fill 96.6\%)} \quad (12)$$

[STATUS: **DERIVED**] C = 7.73 is a numerical consequence. Link to 8 layers is TESTABLE.

§7. Synthesis: ε-Halo as Unified Dark Sector

Scale	Mechanism	Observable
Quark (fm)	$m_d/m_u = 2e^A = 2.1668$	PDG: 2.16±0.08 (0.31%)
Galactic (kpc)	ε-Halo: $\rho \propto 1/r^2$ from θ-gradient	Flat rotation curves
LSS (Mpc)	$S_8 \approx 0.777$, $\Omega_m^{eff} = 0.2908$	DES: 0.776±0.017 (0.06σ)
Cosmic (Gpc)	$\Omega_\Lambda/\Omega_m = 2e^A = 2.1668$	Planck: 2.1746 (0.36%)
i-Tetration	$\eta_{topo} = \lvert z^*\rvert^2 = 0.3221$	$\Omega_m = 0.3140$ (face counting)

No dark matter particles. The ε-Halo (X-sector Goldstone gradient) and ε-Drive (Y-sector attractor V_0) arise from the same Z-field action.

§8. Falsification Registry

Multi-layer structure: [MATH] mathematical (F-A5.1, F-A5.8); [OBS] observational (F-A5.2–A5.7). F-A5.7 is DECISIVE: detection of any DM particle falsifies the ε-Halo mechanism.

ID	Condition	Experiment	Timeline
F-A5.1 [MATH]	$\eta_{topo} \neq \lvert z^*\rvert^2$ (math error)	Analytic check	PROVEN
F-A5.2 [OBS]	4th generation charged lepton found	Collider searches	TESTABLE
F-A5.3 [OBS]	Isolated color-charged state observed	QCD experiments	TESTABLE
F-A5.4 [OBS]	m_d/m_u outside $2e^A$ ± 5% at >3σ	Lattice QCD	TESTABLE
F-A5.5 [OBS]	Ω_m outside 38/121 ± 5% at >5σ	CMB/LSS surveys	TESTABLE
F-A5.6 [OBS]	All galaxies require cuspy profiles	Galaxy surveys	TESTABLE
F-A5.7 [OBS]	DM particles detected (WIMP/axion/etc.)	Direct detection	DECISIVE
F-A5.8 [MATH]	Another (m,k_W) with $\lvert f'\rvert<1$ found	Mathematical proof	PROVEN

F-A5.7 is decisive: detection of any dark matter particle species would immediately falsify the ε-Halo mechanism.

§9. Conclusions

Three convergent routes. i-Tetration ($\eta_{topo} = 0.3221$), face counting ($\Omega_m = 38/121 = 0.3140$), and duality ($1/(1+2e^A) = 0.3157$) all yield $\Omega_m \sim 0.32$ from **A** = 35/437 alone.

ε-Halo mechanism. Massless Goldstone θ-mode satisfies $\Box\theta = 0$ exactly. Gradient energy produces isothermal halos ($\rho \propto 1/r^2$) and flat rotation curves without dark matter particles.

Matter genesis. Stability threshold S = 0.8915 generates three-generation limit ($k_{G,max} = 1.36$). Color-phase pinch (Tension = 1.030 > 1) produces confinement. Mass exponent uses **A** (holonomy), not $\sqrt{3}\cdot\mathbf{A}$ (pinch).

Phase budget. $\varphi(35) = 24 = 4 \times Y = 2 \times G$ encodes spatial–temporal phase structure. Capacity C = 7.73 (≈96.6% of 8-layer closure) is TESTABLE.

Decisive test. Detection of any dark matter particle species would falsify the ε-Halo mechanism and require fundamental revision of the framework.

Acknowledgements & Code Availability

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Code Availability. Verification script: ZS_A5_v1_0_verification.py. Dependencies: Python 3.10+, NumPy. Execution: python3 ZS_A5_v1_0_verification.py. Expected output: 50/50 PASS, exit code 0. Test composition: 47 computational, 1 structural, 2 declarative (4%).

Appendix A. Key Formulae

$z^* = (2i/\pi)\, W_0(-i\pi/2) = 0.4383 + 0.3606i$
$\eta_{topo} = |z^*|^2 = 0.3221$, $S = |z^*|\cdot(\pi/2) = 0.8915$
$\Omega_m = 38/121 = 0.3140$ (face counting, ZS-F2 v1.0 §11)
$\Omega_\Lambda/\Omega_m = m_d/m_u = 2e^A = 2.1668$
$\mathrm{Pinch}(\mathbf{A}, 120°) = 4\sin(\mathbf{A}/2)\sin(60°) = 0.1387$
$k_{G,max} = (1-S)/\mathbf{A} = 1.355 \rightarrow 3$ generations
$\mathrm{Tension}_q = S + \sqrt{3}\cdot\mathbf{A} = 1.030 > 1$ (confinement)
$\varphi(35) = 24$, $C = 24\times0.3221 = 7.73$

Appendix B. Verification Suite Results

Category	Tests	Pass/Fail	Key Result

[A] Locked Inputs	6	6/0	A, (Z,X,Y), z*, φ(35), S, $2e^A$
[B] η_topo & DM Density	5	5/0	η_topo=0.3221, Ω_m(face)=0.3140
[C] ε-Halo Framework	5	5/0	□θ=0, ρ∝1/r², v=const, $a_0=cH_0/6$
[D] Cosmic Budget	5	5/0	Ω_b, Ω_c/Ω_b, Ω_m, duality (face counting)
[E] Phase Budget	4	4/0	φ(5)=4, φ(7)=6, C=7.73
[F] Matter Genesis	5	5/0	Pinch, 3 gen, confinement, Model-E
[G] Branch Stability	4	4/0	k_W=0 unique, k_W=1,2 repulsive
[H] Falsification Gates	8	8/0	F-A5.1–A5.8
[I] Anti-Numerology	3	3/0	0.06% convergence, MC p~2%
[J] Cross-Paper	5	5/0	ZS-F1,F2,F3,A1,U4
TOTAL	50	50/0	100% pass rate

Appendix C. Cross-Reference Table

Result	Status	Dependencies
η_topo = |z*|² = 0.3221	DERIVED	ZS-F3 v1.0 (z* from k_W=0)
Ω_m = 38/121 = 0.3140	DERIVED	ZS-F2 v1.0 §11 (face counting)
Ω_Λ/Ω_m = m_d/m_u = $2e^A$	DERIVED	ZS-F2 v1.0 (A), ZS-A1 v1.0 (duality)
ε-Halo: ρ ∝ 1/r²	DERIVED	ZS-F1 v1.0 (U(1)), ZS-A1 v1.0 (Goldstone θ)
3-generation limit	TESTABLE	ZS-F3 v1.0 (S), ZS-F2 v1.0 (A)
Confinement: Tension > 1	TESTABLE	ZS-F2 v1.0 (A), ZS-F3 v1.0 (S), Pinch
φ(35) = 24 = 4×Y	PROVEN	Number theory + ZS-F5 v1.0
C = 7.73 ≈ 8	DERIVED / TESTABLE	ZS-F3 v1.0 (η_topo), φ(35)

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Version History

v1.0 (March 2026): Initial public release. Consolidated from internal Z-Spin Collaboration research notes up to v2.2.0. Cosmic budget updated to face counting (ZS-F2 v1.0 §11): Ω_m = 38/121 = 0.3140 (was

39/121 slot counting). η_{topo} vs face counting gap (2.5%) noted as under investigation. $S_8 = 0.777$, $\Omega_m^{eff} = 0.2908$. All cross-references use Grand Reset v1.0 codes. Verification: 50/50 PASS.